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Lab 3

Hunting In Memory

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Introduction

This lab involved using the Volatility Framework to conduct a forensic analysis of a Windows XP memory image. Memory forensics is an essential part of Digital Forensics and Incident Response. It allows analysts to access in-memory artifacts that reveal active processes, malware, user activity, and attacker behaviors that may not be captured in other types of analysis. The goal of this lab is to examine the memory dump and identify evidence of compromise, including malicious processes, persistence mechanisms, suspicious file paths, cleartext credentials, and attacker tooling.

The following report documents the methodology used, the forensic findings, and a reconstructed narrative of what occurred on the compromised host.

Methodology

For this lab, all analysis was performed in the win-hunt-09 Simspace VM. Volatility 2.6 was the main tool used within the controlled forensic environment. The *KobayashiMaru.vmem* file was placed into a working directory, and a Volatility alias was created to streamline commands:

- `doskey pro=volatility.exe -f KobayashiMaru.vmem --profile=WinXPSP2x86 $*`

See [Figure 1.1](#) in the Methodology appendix for a screenshot of the Volatility alias. This allowed rapid execution of Volatility plugins using the command prefix `pro`.

Key Volatility Plugins Used

- **imageinfo** - determine OS profile and memory metadata
- **pslist** / **pslist** - enumerate active processes
- **cmdline** - extract command-line arguments used at runtime
- **dlllist** - enumerate loaded DLLs per process

Findings

1. OS & System Information

For step 1, I used the imageinfo plugin to determine the following:

- Operating System: Windows XP
- Architecture: x86 (32-bit)

- Suggested Profile: WinXPSP2x86, WinXPSP3x86
- PAE: No PAE
- Processors: 1
- Memory Timestamp:
 - UTC: 2018-10-30 20:47:03
 - Local: 2018-10-30 14:47:03 (UTC-6)
- RAM Size: Approximately 1 GB (common configuration for XP analysis images).

See [Figure 2.1](#) in the OS and Systems Findings appendix for a screenshot of the imageinfo output. These findings confirm that the target machine is a legacy Windows XP system.

2. Processes

For step 3, I ran *pro pstree* to reveal active processes. Examining the process tree revealed several clearly malicious processes running at the time of the memory capture.

- **poisonivy.exe** - PID 480
 - A known Remote Access Trojan (RAT)
 - Provides attackers with remote desktop access, keylogging, screenshot capture, credential theft, and persistence
 - Never present on a legitimate Windows XP installation
 - Launched under explorer.exe, indicating execution in the logged-in user's session
- **nc.exe (Netcat)** - PID 532
 - Well-known hacker utility used for remote shells and data transfer
 - Should never be present on a standard Windows system
 - Command-line arguments show it was used as a persistent shell listener
- **cmd.exe** - PID 560
 - Legitimate Windows program but suspicious in this context
 - Spawned alongside malware and backdoors
 - Executed with attacker-controlled parameters

See [Figure 3.1](#) in the Processes Appendix for a screenshot of the pstree output. These malicious processes form a clear malware toolchain: Poison Ivy for remote access, Netcat for backdoor shell access, and cmd.exe for staging attacker commands.

3. User Credentials Found in Memory

Using Notepad++ and performing keyword searches on the KobayashiMaru.vmem file revealed the following clear-text credentials.

- Password: B@ji0220! (See [Figure 4.1](#) in the User Credentials Appendix for a screenshot of the password search result)
- User account referenced: Daniel Faraday (See [Figure 4.2](#) in the User Credentials Appendix for a screenshot of the username search result)

This search demonstrated how volatile memory can contain sensitive authentication data that can be extracted by attackers and forensic analysts alike.

4. DLL Analysis

For step 5, I used the dlllist plugin and piped the output to a text file (*pro dlllist > dlllist.txt*). This allowed me to look for the already identified suspicious processes, and analyze the DLLs associated with them.

1. poisonivy.exe (PID 480)

- The DLL list shows the Windows XP libraries associated with the process:
 - kernel32.dll
 - ntdll.dll
 - advapi32.dll
 - ws2_32.dll
 - user32.dll
 - msvcrt.dll
 - ole32.dll
 - version.dll
 - wsock32.dll
 - Rasadhlp.dll
- See [Figure 5.1](#) for a screenshot of the full DLL list for poisonivy.exe.
- Poison Ivy is a known Remote Access Trojan (RAT), so the presence of networking DLLs (ws2_32.dll, wsock32.dll, rasadhlp.dll) is concerning. It confirms that it operates as a network-enabled backdoor, granting the attacker remote control.

2. nc.exe (PID 532)

- Netcat loaded a very large number of DLLs, significantly more than a typical process.
- See [Figure 5.2](#) for a screenshot of the full DLL list for nc.exe.
- One of the most notable DLLs associated with this process is ws2_32.dll. This is a networking DLL used by the legitimate Windows service, Winsock. However, it should never be associated with nc.exe on a normal Windows system. This confirms that this process is an additional form of malware used to backdoor the system.

3. cmd.exe (PID 560)

- The DLL list for cmd.exe shows only standard Windows system DLLs (ntdll.dll, kernel32.dll, user32.dll, etc.). See [Figure 5.3](#) for a screenshot of the full DLL list for cmd.exe.
- However, this is expected because the shell itself is not malware. It is being misused by the attacker. Attackers commonly use cmd.exe after gaining RAT access for post-exploitation actions and further system manipulation.

5. Command-Line Analysis

In addition to the DLL listings, analysis of the command-line output (pro cmdline) revealed multiple abnormal, high-risk file paths that clearly indicate attacker activity. These paths do not correspond to legitimate Windows XP system locations and are commonly used by threat actors for staging malware, persistence, and backdoor tools.

1. poisonivy.exe (PID 480)

- **Command line:** "C:\WINDOWS\system32\poisonivy.exe"
- The RAT was deliberately placed in System32, a common evasion technique that helps it blend into legitimate system binaries.
- See [Figure 6.3](#) for a screenshot of poisonivy.exe in the pro cmdline output.

2. nc.exe (PID 532)

- **Command line:** "C:\inetpub\ftproot\nc.exe -L -p 6666 -e cmd.exe"
 - "-L" means it is a persistent listener, likely used by the attacker to re-establish shells on every connection.

- “-p 6666” refers to port 6666. A non-standard port typically used for stealth.
- “-e cmd.exe” binds cmd.exe to a socket. This provides the attacker with a fully remote command shell.

3. cmd.exe (PID 560)

- **Command line:** “C:\WINDOWS\system32\cmd.exe /K C:\inetpub\ftproot\lock.bat”
 - “/K” keeps the shell open, indicating interactive use from the attacker.
 - “lock.bat” is stored inside the compromised FTP root and is used for further system manipulation.
 - This reflects the typical scripted automation of attacker activities.

Incident Explanation

The memory analysis indicates that the victim system suffered a significant malware compromise initiated by the Poison Ivy Remote Access Trojan. Once executed, Poison Ivy allowed the attacker to control the system remotely, resulting in the installation of multiple backdoors and hacking utilities.

The attacker went on to place and execute nc.exe (Netcat) in C:\inetpub\ftproot\. This opened a persistent command shell on port 6666. With this backdoor, the attacker enabled remote execution of Windows commands. This resulted in the attacker gaining full control of the victim machine. The attacker also executed a batch script (lock.bat) through cmd.exe /K, demonstrating automation and further manipulation of the environment.

Additionally, user credentials were found in the memory (password: B@ji0220! and account "Daniel Faraday"). This confirms that sensitive data was exposed and accessible to the attacker. The combination of RATs, backdoor listeners, malicious scripts, and credentials confirms that the system was actively under attacker control at the time of capture.

Conclusion

The analysis conducted in this lab revealed how powerful memory forensics is in uncovering active attacker activity that might otherwise go unnoticed. The use of Volatility allowed the extraction of detailed information about running processes, command-line arguments, loaded DLLs, and malware paths. These insights enabled the

reconstruction of a full attack chain. This analysis emphasizes the importance of memory forensics in real-world digital investigations and illustrates how DFIR analysts identify and understand attacker behavior.

References

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Appendix

1. Methodology

- Figure 1.1: Screenshot of the Volatility Alias.

```
C:\Users\Administrator\Desktop\Lab3>doskey pro=volatility.exe -f KobayashiMaru.vmem $*

C:\Users\Administrator\Desktop\Lab3>pro imageinfo
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
      Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
      AS Layer1 : IA32PagedMemory (Kernel AS)
      AS Layer2 : FileAddressSpace (C:\Users\Administrator\Desktop\Lab3\KobayashiMaru.vmem)
      PAE type : No PAE
      DTB : 0x39000L
      KDBG : 0x80537d60L
      Number of Processors : 1
      Image Type (Service Pack) : 0
      KPCR for CPU 0 : 0xffdff000L
      KUSER_SHARED_DATA : 0xffdf0000L
      Image date and time : 2018-10-30 20:47:03 UTC+0000
      Image local date and time : 2018-10-30 14:47:03 -0600
```

2. OS and System Findings

- Figure 2.1: Screenshot of the imageinfo Output

```
C:\Users\Administrator\Desktop\Lab3>volatility -f KobayashiMaru.vmem imageinfo
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
      Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
      AS Layer1 : IA32PagedMemory (Kernel AS)
      AS Layer2 : FileAddressSpace (C:\Users\Administrator\Desktop\Lab3\KobayashiMaru.vmem)
      PAE type : No PAE
      DTB : 0x39000L
      KDBG : 0x80537d60L
      Number of Processors : 1
      Image Type (Service Pack) : 0
      KPCR for CPU 0 : 0xffdff000L
      KUSER_SHARED_DATA : 0xffdf0000L
      Image date and time : 2018-10-30 20:47:03 UTC+0000
      Image local date and time : 2018-10-30 14:47:03 -0600
```

3. Processes

- Figure 3.1: Screenshot of pstree Output

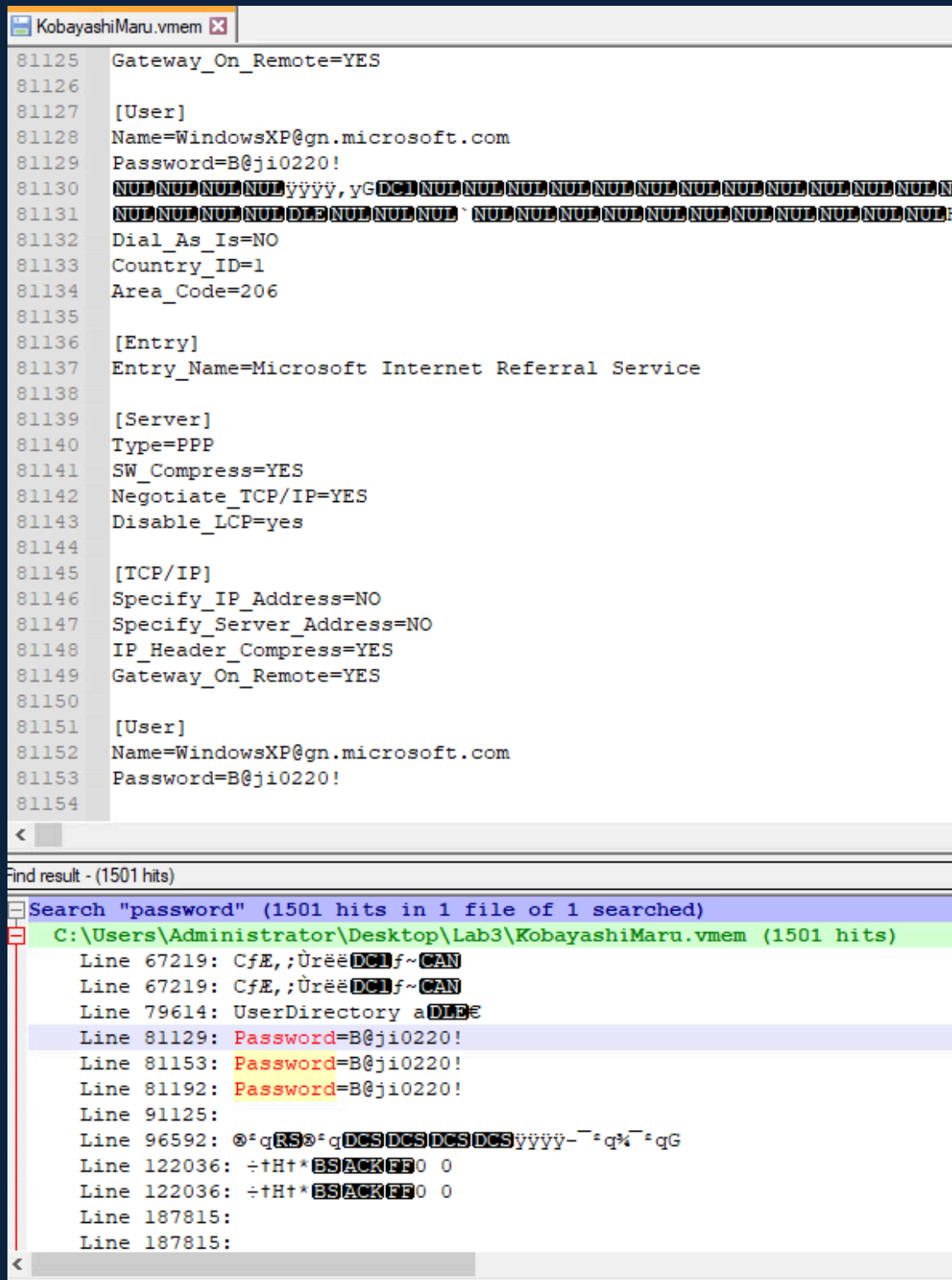
```
C:\Users\Administrator\Desktop\Lab3>pro pstree
Volatility Foundation Volatility Framework 2.6
```

Name	Pid	PPid	Thds	Hnds	Time

0x81fcc800:System	4	0	54	275	1970-01-01 00:00:00 UTC+0000
. 0x81f07da8:smss.exe	336	4	3	21	2018-10-30 20:46:44 UTC+0000
.. 0x81d2b020:csrss.exe	664	336	12	453	2018-10-30 20:46:45 UTC+0000
.. 0x81dc4020:winlogon.exe	688	336	25	486	2018-10-30 20:46:45 UTC+0000
... 0x819efda8:services.exe	732	688	18	390	2018-10-30 20:46:45 UTC+0000
.... 0x81d626a0:inetinfo.exe	1432	732	34	540	2018-10-30 20:46:46 UTC+0000
.... 0x81db4298:hxddef100.exe	1416	732	2	31	2018-10-30 20:46:46 UTC+0000
..... 0x81ede980:cryptcat.exe	1472	1416	1	62	2018-10-30 20:46:47 UTC+0000
..... 0x81cada80:bircd.exe	1480	1416	2	45	2018-10-30 20:46:47 UTC+0000
.... 0x81d32988:wmiapsrv.exe	216	732	5	121	2018-10-30 20:46:36 UTC+0000
.... 0x819edda8:svchost.exe	916	732	9	252	2018-10-30 20:46:45 UTC+0000
..... 0x819e83c8:wmiprvse.exe	252	916	7	107	2018-10-30 20:46:37 UTC+0000
.... 0x81d976c8:svchost.exe	1028	732	5	72	2018-10-30 20:46:45 UTC+0000
.... 0x81e536a0:spoolsv.exe	1308	732	15	189	2018-10-30 20:46:46 UTC+0000
.... 0x819e2c20:jqc.exe	1464	732	7	214	2018-10-30 20:46:47 UTC+0000
.... 0x81ee5500:svchost.exe	960	732	70	875	2018-10-30 20:46:45 UTC+0000
.... 0x81e07da8:svchost.exe	1108	732	12	142	2018-10-30 20:46:46 UTC+0000
.... 0x81c71508:VMwareService.e	1624	732	2	119	2018-10-30 20:46:47 UTC+0000
.... 0x81e92418:vmacthlp.exe	888	732	1	27	2018-10-30 20:46:45 UTC+0000
.... 0x81b98da8:lsass.exe	744	688	25	339	2018-10-30 20:46:45 UTC+0000
... 0x81b82638:logonui.exe	636	688	4	133	2018-10-30 20:46:40 UTC+0000
... 0x81edfc18:userinit.exe	368	688	2	34	2018-10-30 20:46:38 UTC+0000
.... 0x81a3bc18:explorer.exe	404	368	15	252	2018-10-30 20:46:38 UTC+0000
..... 0x81d28790:VMwareTray.exe	456	404	1	30	2018-10-30 20:46:38 UTC+0000
..... 0x81e234e8:poisonivy.exe	480	404	1	20	2018-10-30 20:46:38 UTC+0000
..... 0x81bb3da8:VMwareUser.exe	464	404	5	146	2018-10-30 20:46:38 UTC+0000
..... 0x81aaa708:jusched.exe	472	404	1	24	2018-10-30 20:46:38 UTC+0000
..... 0x81cacda8:msmsgs.exe	488	404	4	127	2018-10-30 20:46:39 UTC+0000
..... 0x81d40418:rundll32.exe	984	404	1	81	2018-10-30 20:46:43 UTC+0000
0x81e579f8:soffice.exe	516	496	1	20	2018-10-30 20:46:39 UTC+0000
. 0x81ec6848:soffice.bin	524	516	7	164	2018-10-30 20:46:39 UTC+0000
0x81e8f9c0:iroffer.exe	1692	1488	0	-----	2018-10-30 20:46:47 UTC+0000
. 0x81c85420:iroffer.exe	1728	1692	5	92	2018-10-30 20:46:47 UTC+0000
.. 0x81df6b20:iroffer.exe	1824	1728	0	-----	2018-10-30 20:46:47 UTC+0000
0x81a2eb78:cmd.exe	560	508	1	20	2018-10-30 20:46:39 UTC+0000
0x81eb3020:winvnc4.exe	548	508	2	81	2018-10-30 20:46:39 UTC+0000
0x81c6f7b8:nc.exe	532	508	1	62	2018-10-30 20:46:39 UTC+0000

4. User Credentials

- Figure 4.1: Screenshot of the Password Search Result



```

81125 Gateway_On_Remote=YES
81126
81127 [User]
81128 Name=WindowsXP@gn.microsoft.com
81129 Password=B@ji0220!
81130 NUL NUL NUL NUL yyy, yG DCS NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL N
81131 NUL NUL NUL NUL DLE NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL NUL N
81132 Dial_As_Is=NO
81133 Country_ID=1
81134 Area_Code=206
81135
81136 [Entry]
81137 Entry_Name=Microsoft Internet Referral Service
81138
81139 [Server]
81140 Type=PPP
81141 SW_Compress=YES
81142 Negotiate_TCP/IP=YES
81143 Disable_LCP=yes
81144
81145 [TCP/IP]
81146 Specify_IP_Address=NO
81147 Specify_Server_Address=NO
81148 IP_Header_Compress=YES
81149 Gateway_On_Remote=YES
81150
81151 [User]
81152 Name=WindowsXP@gn.microsoft.com
81153 Password=B@ji0220!
81154

```

Find result - (1501 hits)

Search "password" (1501 hits in 1 file of 1 searched)

C:\Users\Administrator\Desktop\Lab3\KobayashiMaru.vmem (1501 hits)

```

Line 67219: CfE,;ÜrëëDCI,f~CAN
Line 67219: CfE,;ÜrëëDCI,f~CAN
Line 79614: UserDirectory aDLE€
Line 81129: Password=B@ji0220!
Line 81153: Password=B@ji0220!
Line 81192: Password=B@ji0220!
Line 91125:
Line 96592: @qRS@qDCSDCSDCSDCSyyy~zq%zqG
Line 122036: ÷tH+*BSACKFF0 0
Line 122036: ÷tH+*BSACKFF0 0
Line 187815:
Line 187815:

```

- **Figure 4.2: Screenshot of the Username Search Result**

```
KobayashiMaru.vmem
78894 }ÏÊSOHBSUSéÊENQêÊSOH@ > & VtETXdesktop.ini X
78895 LocalizedResourceName=@shell32.dll,-21774
78896 ŷŷŷŷ,yGDC1
78897   'p SOH-EM  'p SOHBSUS   ENQ   SOH jSOH 0 p
78898 |'p SOH"[
78899 |'p SOH"[
78900 |'p SOH DC1SOH 0 h STX L CAN SOH ES
78901 Owner=Daniel Faraday
78902 Personalized=5
78903 PersonalizedName=My Documents
78904 ŷŷŷŷ,yGDC1
78905 Owner=Daniel Faraday
78906 Personalized=39
78907 PersonalizedName=My Pictures
78908 [.ShellClassInfo]
78909 InfoTip=@Shell32.dll,-12688
78910 IconFile=%SystemRoot%\System32\mydocs.dll
78911 IconIndex=-101
78912 ŷŷŷŷ,yGDC1
78913 Owner=Daniel Faraday
78914 Personalized=13
78915 PersonalizedName=My Music
78916 [.ShellClassInfo]
78917 InfoTip=@Shell32.dll,-12689
78918 IconFile=%SystemRoot%\system32\shell32.dll
78919 IconIndex=-237
78920 ŷŷŷŷ,yGDC1
<
Find result - (647 hits)
Search "owner" (647 hits in 1 file of 1 searched)
C:\Users\Administrator\Desktop\Lab3\KobayashiMaru.vmem (647 hits)
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9204: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
Line 9282: THREAD_NOT_MUTEX OWNER
Line 10767: SO ST DLE DC1 DC2 P Q R S T U V W X Y Z [ HAL.dll ExAcquireFastMutex ExRel
Line 78901: Owner=Daniel Faraday
Line 78905: Owner=Daniel Faraday
Line 78913: Owner=Daniel Faraday
Line 87658: STXKeTickCount ntoskrnl.exe R KfReleaseSpinLock O KfAcquireSpinLock W READ_P
Line 89742: ENQSOENQSIENQDLEENQDC1ENQDC2ENQDC3ENQDC4ENQNAKENQSYNENQETBENQCANENQEM
<
Normal text file
```

5. DLLs Associated with Suspicious Processes

- Figure 5.1: Screenshot of DLLs Associated with *poisonivy.exe*.

```
poisonivy.exe pid: 480
Command line : "C:\WINDOWS\System32\poisonivy.exe"
```

Base	Size	LoadCount	Path
0x00400000	0x1c00	0xffff	C:\WINDOWS\System32\poisonivy.exe
0x77f50000	0xa9000	0xffff	C:\WINDOWS\System32\ntdll.dll
0x77e60000	0xe5000	0xffff	C:\WINDOWS\system32\kernel32.dll
0x77dd0000	0x8b000	0x1a	C:\WINDOWS\system32\advapi32.dll
0x77cc0000	0x75000	0xb	C:\WINDOWS\system32\RPCRT4.dll
0x77d40000	0x8d000	0x5	C:\WINDOWS\system32\user32.dll
0x77c70000	0x40000	0x4	C:\WINDOWS\system32\GDI32.dll
0x75260000	0x27000	0x1	C:\WINDOWS\System32\advpack.dll
0x771b0000	0x11a000	0x1	C:\WINDOWS\system32\ole32.dll
0x77c00000	0x7000	0x1	C:\WINDOWS\system32\VERSION.dll
0x71ab0000	0x15000	0x5	C:\WINDOWS\System32\ws2_32.dll
0x77c10000	0x53000	0x8	C:\WINDOWS\system32\msvcrt.dll
0x71aa0000	0x8000	0x7	C:\WINDOWS\System32\WS2HELP.dll
0x71a50000	0x3b000	0x2	C:\WINDOWS\system32\mswsock.dll
0x71a90000	0x8000	0x1	C:\WINDOWS\System32\wshtcpip.dll
0x76fc0000	0x5000	0x1	C:\WINDOWS\System32\rasadhlp.dll

- Figure 5.2: Screenshot of DLLs Associated with nc.exe.

dlllist - Notepad

File Edit Format View Help

nc.exe pid: 532

Command line : C:\inetpub\ftproot\nc.exe -L -p 6666 -e cmd.exe

Base	Size	LoadCount	Path
0x00400000	0x10000	0xffff	C:\inetpub\ftproot\nc.exe
0x77f50000	0xa9000	0xffff	C:\WINDOWS\System32\ntdll.dll
0x77e60000	0xe5000	0xffff	C:\WINDOWS\system32\kernel32.dll
0x71ab0000	0x15000	0xffff	C:\WINDOWS\System32\WS2_32.dll
0x77c10000	0x53000	0xffff	C:\WINDOWS\system32\msvcrt.dll
0x71aa0000	0x8000	0xffff	C:\WINDOWS\System32\WS2HELP.dll
0x77dd0000	0x8b000	0xffff	C:\WINDOWS\system32\ADVAPI32.dll
0x77cc0000	0x75000	0xffff	C:\WINDOWS\system32\RPCRT4.dll
0x71a50000	0x3b000	0x4	C:\WINDOWS\System32\mswsock.dll
0x76f20000	0x25000	0x3	C:\WINDOWS\System32\DNSAPI.dll
0x76d60000	0x15000	0x3	C:\WINDOWS\System32\iphlpapi.dll
0x76de0000	0x26000	0x1	C:\WINDOWS\System32\netman.dll
0x76d40000	0x16000	0x1	C:\WINDOWS\System32\PPRAPI.dll
0x76e40000	0x2f000	0x1	C:\WINDOWS\System32\ACTIVEDS.dll
0x76e10000	0x24000	0x1	C:\WINDOWS\System32\adsldpc.dll
0x71c20000	0x4f000	0x6	C:\WINDOWS\System32\NETAPI32.dll
0x76f60000	0x2c000	0x2	C:\WINDOWS\system32\WLDAP32.dll
0x77d40000	0x8d000	0x28	C:\WINDOWS\system32\USER32.dll
0x77c70000	0x40000	0x16	C:\WINDOWS\system32\GDI32.dll
0x76b20000	0x15000	0x1	C:\WINDOWS\System32\ATL.DLL
0x771b0000	0x11a000	0x7	C:\WINDOWS\system32\ole32.dll
0x77120000	0x8b000	0x4	C:\WINDOWS\system32\OLEAUT32.dll
0x76e80000	0xd000	0x4	C:\WINDOWS\System32\rtutils.dll
0x71bf0000	0x11000	0x1	C:\WINDOWS\System32\SAMLIB.dll
0x76670000	0xe4000	0x1	C:\WINDOWS\System32\SETUPAPI.dll
0x76ee0000	0x37000	0x2	C:\WINDOWS\System32\RASAPI32.dll
0x76e90000	0x11000	0x2	C:\WINDOWS\System32\rasman.dll
0x76eb0000	0x2a000	0x2	C:\WINDOWS\System32\TAPI32.dll
0x772d0000	0x63000	0x6	C:\WINDOWS\system32\SHLWAPI.dll
0x76b40000	0x2c000	0x2	C:\WINDOWS\System32\WINMM.dll
0x773d0000	0x7f4000	0x1	C:\WINDOWS\system32\SHELL32.dll
0x76f90000	0x10000	0x3	C:\WINDOWS\System32\Secur32.dll
0x76da0000	0x30000	0x1	C:\WINDOWS\System32\WZCSvc.DLL
0x76d30000	0x4000	0x1	C:\WINDOWS\System32\WMFI.dll
0x76d80000	0x1a000	0x1	C:\WINDOWS\System32\DHCPSCVC.DLL
0x762c0000	0x8a000	0x1	C:\WINDOWS\system32\CRYPT32.dll
0x762a0000	0xf000	0x1	C:\WINDOWS\system32\VSASNI.dll
0x76f50000	0x8000	0x1	C:\WINDOWS\System32\WTSAPI32.dll
0x76360000	0xf000	0x2	C:\WINDOWS\System32\WINSTA.dll
0x71950000	0xe4000	0x2	C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.0.0_x-ww_1382d70a\comctl32.dll
0x77340000	0x8b000	0x1	C:\WINDOWS\system32\comctl32.dll
0x76fb0000	0x7000	0x1	C:\WINDOWS\System32\winnrn.dll
0x76fc0000	0x5000	0x1	C:\WINDOWS\System32\rasadhlp.dll
0x71a90000	0x8000	0x1	C:\WINDOWS\System32\wshtcpip.dll

- Figure 5.3: Screenshot of DLLs Associated with cmd.exe.

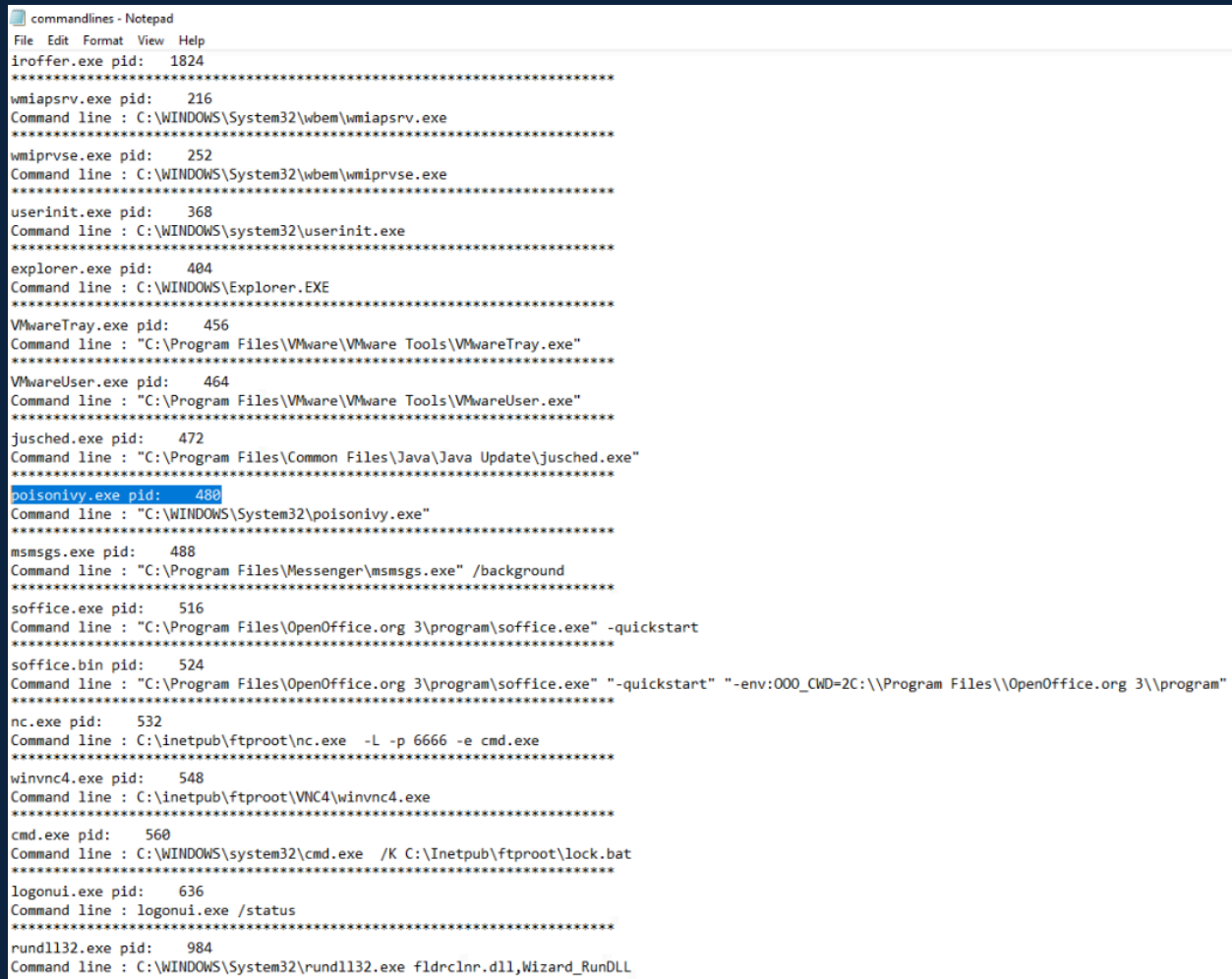
cmd.exe pid: 560

Command line : C:\WINDOWS\system32\cmd.exe /K C:\Inetpub\ftproot\lock.bat

Base	Size	LoadCount	Path
0x4ad00000	0x5e000	0xffff	C:\WINDOWS\system32\cmd.exe
0x77f50000	0xa9000	0xffff	C:\WINDOWS\System32\ntdll.dll
0x77e60000	0xe5000	0xffff	C:\WINDOWS\system32\kernel32.dll
0x77c10000	0x53000	0xffff	C:\WINDOWS\system32\msvcrt.dll
0x77d40000	0x8d000	0xffff	C:\WINDOWS\system32\USER32.dll
0x77c70000	0x40000	0xffff	C:\WINDOWS\system32\GDI32.dll
0x77dd0000	0x8b000	0xffff	C:\WINDOWS\system32\ADVAPI32.dll
0x77cc0000	0x75000	0xffff	C:\WINDOWS\system32\RPCRT4.dll

6. Command Line Findings

- Figure 6.3: Screenshot of *poisonivy.exe* in the pro cmdline Output.



```
commandlines - Notepad
File Edit Format View Help
iroffer.exe pid: 1824
*****
wmiapsrv.exe pid: 216
Command line : C:\WINDOWS\System32\wbem\wmiapsrv.exe
*****
wmiprvse.exe pid: 252
Command line : C:\WINDOWS\System32\wbem\wmiprvse.exe
*****
userinit.exe pid: 368
Command line : C:\WINDOWS\system32\userinit.exe
*****
explorer.exe pid: 404
Command line : C:\WINDOWS\Explorer.EXE
*****
VMwareTray.exe pid: 456
Command line : "C:\Program Files\VMware\VMware Tools\VMwareTray.exe"
*****
VMwareUser.exe pid: 464
Command line : "C:\Program Files\VMware\VMware Tools\VMwareUser.exe"
*****
jusched.exe pid: 472
Command line : "C:\Program Files\Common Files\Java\Java Update\jusched.exe"
*****
poisonivy.exe pid: 480
Command line : "C:\WINDOWS\System32\poisonivy.exe"
*****
msmsgs.exe pid: 488
Command line : "C:\Program Files\Messenger\msmsgs.exe" /background
*****
soffice.exe pid: 516
Command line : "C:\Program Files\OpenOffice.org 3\program\soffice.exe" -quickstart
*****
soffice.bin pid: 524
Command line : "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-quickstart" "-env:000_CWD=2C:\\Program Files\\OpenOffice.org 3\\program"
*****
nc.exe pid: 532
Command line : C:\inetpub\ftproot\nc.exe -L -p 6666 -e cmd.exe
*****
winvnc4.exe pid: 548
Command line : C:\inetpub\ftproot\VNC4\winvnc4.exe
*****
cmd.exe pid: 560
Command line : C:\WINDOWS\system32\cmd.exe /K C:\Inetpub\ftproot\lock.bat
*****
logonui.exe pid: 636
Command line : logonui.exe /status
*****
rundll32.exe pid: 984
Command line : C:\WINDOWS\System32\rundll32.exe fldrclnr.dll,Wizard_RunDLL
```